(Please write your Enrollment Number) Enrollment No. 02701022022

End-TermExamination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:B.Tech, Semester: 3
(May, 2024)

Subject Code: BEC 206	Subject: Electromagnetic Field Theory

Time :3 Hours Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1			**(2.5*8=20)**	
	(a)	Explain equipotential surfaces with their significance.		
	(b)	Explain divergence theorem and evaluate the dot product of the vectors 2i + 5j – 3k and–i + 4j – 2k.		
	(c)	Derive the relation between electric field intensity and current density.		
	(d)	Derive electric field on the point 'a' meter apart laterally from the end of the uniform linear continuous charge distribution.		
	(e)	State ampere's circuit Law.		
	(f)	Explain continuity equation and define relaxation time.		
	(g)	An air line has a characteristic impedance of 70Ω and a phase constant of 3rad/m at 100MHz. Calculate the inductance per meter and capacitance per meter of the line.		
	(h)	What is skin depth. Derive its relation with attenuation constant.		
UNIT-I				
Q2	Given the electric field intensity, E = (10/r2)ar+$sin\theta cos\emptyset a\theta$+ $2cos2\emptyset a\emptyset$. Find the divergence and curl of E at $(2,\pi/2,0)$.		**(10)**	
Q3	Write the equations for the conversion of spherical coordinates to the cartesian coordinates.		**(10)**	
UNIT-II				
Q4	Demonstrate the Gauss's law with its applications.		**(10)**	
Q5	Derive the boundary conditions for a dielectric dielectric interface.		**(10)**	
UNIT-III				
Q6	(a)Discuss Biot-Savart's Law. (b) A circular loop located on $x^2 + y^2 = 9$, z = 0 carries a direct current of 10 A along aØ. Determine H at (0, 0, 4) and (0, 0, -4).		**(10)**	
Q7	(a)Discuss about Maxwell's field equations in differential and integral andform and explain their significance.		**(10)**	
UNIT-IV				
Q8	(a) State and derive the poynting theorem. (b) In a nonmagnetic medium, $E = 4 \sin (2\pi \times 10^7 t - 0.8x)$ azV/m Find (i) $\in r$, η (j) The time-average power carried by the wave (k) The total power crossing 100 cm^2 of plane 2x + y = 5		**(10)**	
Q9	(a) Explain equivalent circuit of transmission line and highlight the line parameters, characteristic impedance and propagation constant. (b) Discuss lossless and distortion less transmission lines.		**(10)**	